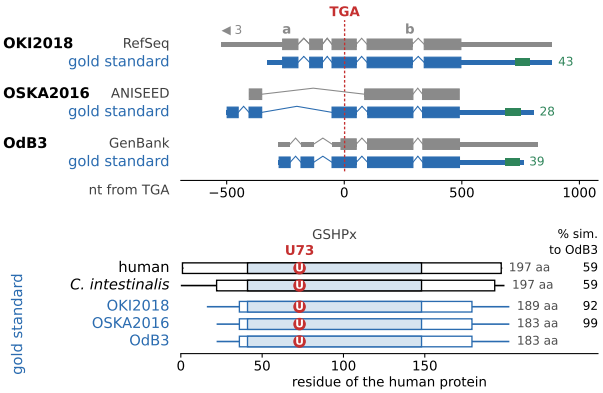
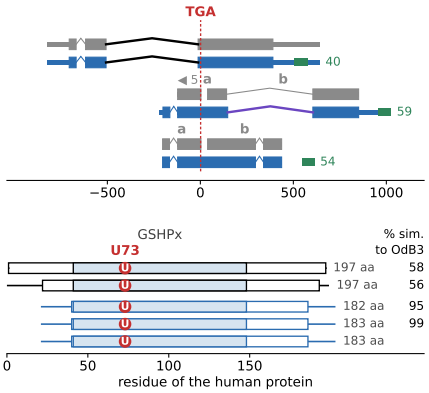
GPX_PAR



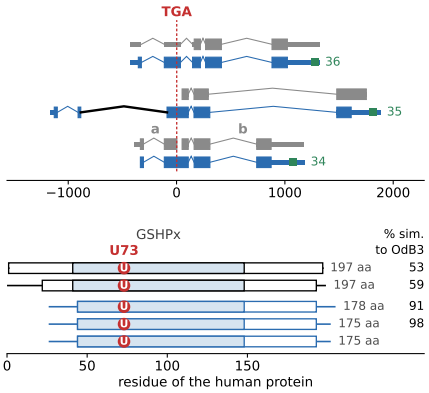
OKI2018 RefSeq a XR_014134294.1: non-coding RNA that starts upstream and ends in the TGA exon; b XP_081635498.1: selenocysteine declared; same model as the gold standard.
OSKA2016 ANISEED S25.g14813.01.t: TGA exon skipped; starts in the second exon.
OdB3 GenBank CBY25160.1: starts 14 nt before the TGA, in another frame for 41 codons. Gold standard: mapped from the transcript (one extra base in the assembly).
proteins human P36969; C. intestinalis Cint_GPX.5.

GPX_chr1a



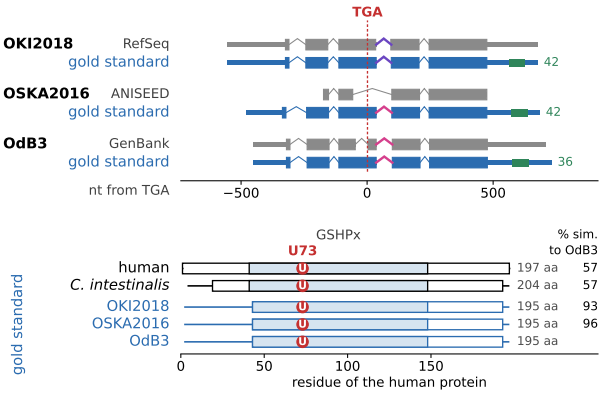
OKI2018 RefSeq XP_081637767.1: same model; TGA translated as X (selenocysteine not declared).
OSKA2016 ANISEED a S6.g07306.01.t: other 5′ exons; TGA read as stop; b S6.g07307.01.t: starts 13 codons after the TGA.
OdB3 GenBank a CBY23346.1: TGA read as stop; b CBY23347.1: starts 13 codons after the TGA. Gold standard: the two GenBank models joined through the TGA (no transcript).
proteins human P36969; C. intestinalis Cint_GPX.5.

GPX_chr1b



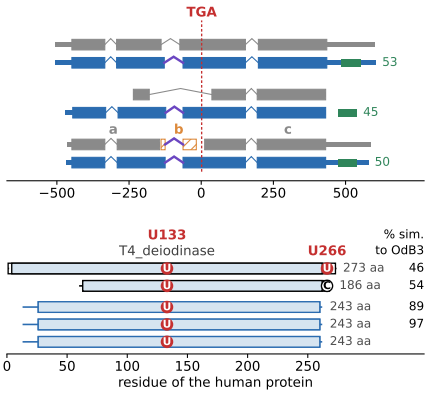
OKI2018 RefSeq XP_081636615.1: same exon chain; coding sequence starts at an internal ATG, 17 codons after the TGA.
OSKA2016 ANISEED S3.g04759.01.t: starts 17 codons after the TGA; reads through one intron, places another differently.
OdB3 GenBank a CBY19528.1: TGA read as stop; b CBY19527.1: starts 17 codons after the TGA.
proteins human P36969; C. intestinalis Cint_GPX.5.

GPX_chr2



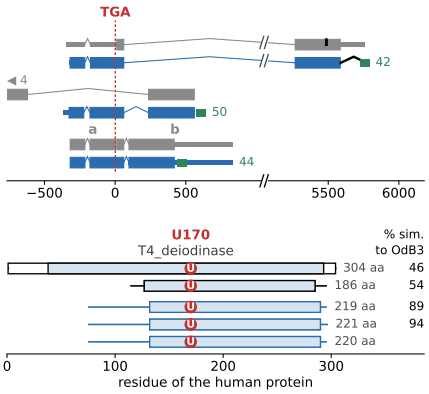
OKI2018 RefSeq XP_081630045.1: same exon chain; TGA translated as X (selenocysteine not declared).
OSKA2016 ANISEED S27.g15046.01.t: starts in the second exon; TGA in an intron of the coding sequence.
OdB3 GenBank CBY22474.1: TGA in an extra intron of the coding sequence.
proteins human P36969; C. intestinalis Cint_GPX.1.

DIO_PAR



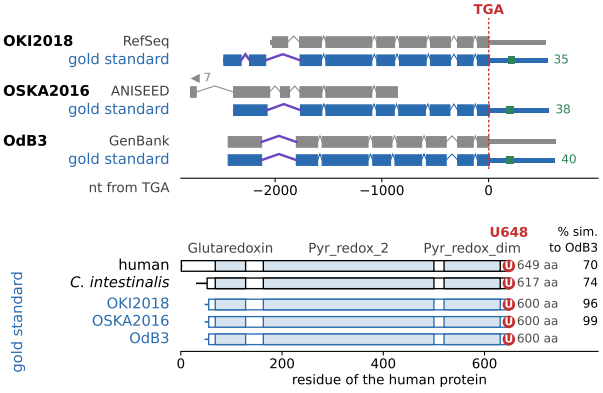
OKI2018 RefSeq XP_081636384.1: selenocysteine declared; one intron 11 nt away from the GA-AG intron of the transcripts.
OSKA2016 ANISEED S2.g03083.01.t: TGA in an intron of the annotated coding sequence.
OdB3 GenBank a CBY14356.1: ends before the TGA; b CBY14355.1: other frame; c CBY14354.1: starts after the TGA. Gold standard: protein from the ESTs (one extra base in the assembly).
proteins human Q92813; C. intestinalis Cint_DIO.2.

DIO_chr1



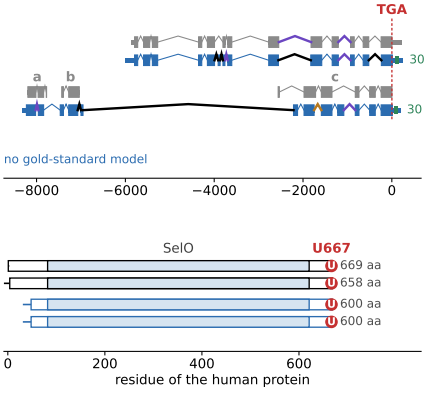
OKI2018 RefSeq XP_081632420.1: coding sequence starts 4 codons after the TGA; model corrected for the assembly (low-quality protein). Gold standard: mapped from the transcript (the assembly lacks 7 bases of the coding sequence and has 6 others).
OSKA2016 ANISEED S37.g16254.01.t: TGA exon skipped; joined to 5 exons upstream of the gene.
OdB3 GenBank a CBY06720.1: ends at the TGA (read as stop); b CBY06719.1: starts 4 codons after the TGA.
proteins human P55073; C. intestinalis Cint_DIO.2.

TXNRD



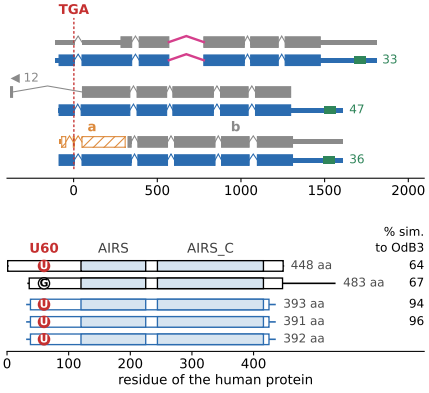
OKI2018 RefSeq XP_081643508.1: N-terminus from other exons (no glutaredoxin domain); TGA read as stop.
OSKA2016 ANISEED S8.g08759.01.t: extra exon in the GA-AG intron; ends in an intron before the TGA.
OdB3 GenBank CBY19735.1: one intron differs from the ESTs; TGA read as stop. Gold standard: exons beyond the ESTs from the GenBank model.
proteins human Q16881; C. intestinalis Cint_TXNRD.14.

SELENOO



OKI2018 RefSeq XP_081634508.1: TGA read as stop; GT-AG or GA-AG introns where the gold standard, mapped from the transcript, has other sites. Gold standard: mapped from the transcript (the assembly lacks 5 bases of the coding sequence).
OSKA2016 ANISEED a S11.g10191.01.t: N-terminal part; b S11.g10192.01.t: central part; c S11.g10193.01.t: C-terminal part, TGA read as stop, skips one exon.
OdB3 no gold-standard model: the exon of the TGA is not in the assembly.
proteins human Q9BVL4; C. intestinalis Cint_SELENOO.1.

SEPHS2



OKI2018 RefSeq XP_081635792.1: coding sequence starts 79 codons after the TGA.
OSKA2016 ANISEED S25.g14713.01.t: TGA exon skipped; fused with the upstream gene.
OdB3 GenBank a CBY08599.1: other frame; b CBY08598.1: starts at an internal ATG.
proteins human Q99611; C. intestinalis Cint_SEPHS2.5.

exon: coding (thick), UTR (thin) — public model — gold-standard model — coding sequence in another frame
selenocysteine TGA (x axis: nucleotides from it, gene orientation) — 37 best SECIS hit and bit score (Oikopleura model; open box: below 20 bits)
◂n number of exons beyond the drawn region — a b public models of the same row, as named in the paragraph
intron splice sites (drawn length unchanged): GT-AG (colour of the model) — GA-AG — GC-AG — GG-AG — other site
Pfam domain of the first reference — U selenocysteine aligned to one of the first reference — © other residue aligned to it
residues not aligned to the first reference — 92 % similar residues to the OdB3 protein (BLOSUM62 > 0; global alignment, aligned part)
human: UniProt; C. intestinalis: stage 1 predictions on its genome (accessions in the paragraphs)